

# Thread & Core

S Y S T E M S

ENTERPRISE ARCHITECTURE FOR THE CONNECTED ENTERPRISE

SAP BTP · Clean Core · Business AI · Integration · Full-Stack Engineering

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CAPABILITY DECK · 2026

Founder-led practice · Pune, India · Remote & onsite, globally

[threadandcore.com](https://threadandcore.com)



# Enterprise software is becoming AI-native.

SAP is embedding business AI across its entire portfolio — Joule as the assistant layer, AI Foundation on BTP as the developer stack. The ERP stays the system of record; BTP is the system of innovation. The winners will be the enterprises that connect AI to process, data, and governance first.

**\$2.6–4.4T**

potential annual value added by generative AI to the global economy

**1 in 3**

companies already use generative AI in at least one business function

**40%**

of organisations are increasing AI investment because of generative AI

Source: McKinsey, State of AI — as presented in SAP's BTP & Generative AI strategy.



## Joule

the assistant across every SAP product



## Embedded AI

AI-powered capabilities inside business processes



## AI Foundation on BTP

AI Core, GenAI Hub, vector engine — where custom AI is built

*We build on this stack today — Joule Studio, SAP AI Core, RAG on HANA vector — in production, not in slideware.*



# Five pillars of SAP BTP. We deliver across all of them.

BTP is how enterprises innovate without touching the ERP core — build, integrate, analyse, automate, and apply AI on one governed platform.



## Application Development

Side-by-side extensions with CAP, Fiori/UI5, SAP Build Apps, and MDK mobile.



## Automation

SAP Build Process Automation — approvals, workflows, document-driven processes.



## Integration

Integration Suite, APIs, events — SAP and non-SAP, with principal propagation.



## Data & Analytics

HANA Cloud, Datasphere, SAP Analytics Cloud — live decision intelligence.



## AI & Machine Learning

SAP AI Core, Joule & Joule Studio, GenAI Hub, vector engine, RAG.

**OUR DEEPEST EXPERTISE**    **BTP Side-by-Side Extensions · BTP Integration · SAP Joule & Joule Studio · SAP AI Core · Conversational AI · Low-Code (SAP Build)**



# Extend the core. Never modify it.

PLANNERS · OPERATORS · CUSTOMERS · MANAGERS

## SIDE-BY-SIDE EXTENSIONS ON SAP BTP

CAP services · Fiori / React UX · MDK mobile · HANA Cloud persistence ·  
AI Core & Joule · Build Process Automation

*All custom logic, data, and UX live here — upgrade-safe.*

released APIs · CDS · events | principal propagation only

## SAP S/4HANA — THE DIGITAL CORE

System of record. Standard processes, standard code — untouched,  
always upgrade-ready.

*No core modifications. Ever.*

## HOW WE KEEP IT CLEAN



**Released APIs only** — Every S/4HANA call uses released OData / CDS interfaces — nothing fragile, nothing modified.



**Right extensibility tier** — In-app for small fits, side-by-side for real applications — decided per use case, documented.



**Identity flows end-to-end** — XSUAA + principal propagation: S/4HANA enforces its own authorisations on every call. No shared technical users.



**Upgrade-safe lifecycle** — Versioned, transportable, CI/CD-deployed — core upgrades never wait for custom code.



**Custom-code triage at scale** — We have classified 45,000+ legacy objects into clean-core tiers to de-risk S/4HANA moves.



# Five services. One architecture.



## ERP Extensions & Integration

Side-by-side applications on a clean ERP core — plus the APIs, events, and middleware that connect them.

SAP BTP · CAP · FIORI/UI5 · INTEGRATION SUITE



## AI & Intelligent Automation

Forecasting, copilots, RAG, and workflow automation — deployed where they remove real work.

SAP AI CORE · JOULE · GENAI · RAG · SAP BPA



## Full-Stack Enterprise Platforms

Web and mobile platforms teams run the business on — offline-first, multilingual, enterprise-secured.

REACT · NODE.JS · MONGODB · MOBILE



## Data & Analytics

Live enterprise data turned into dashboards and decision intelligence — health, visibility, migration readiness.

HANA CLOUD · SAC · DATASPHERE · CAP



## Planning & Solution Architecture

Ecosystem study, DPRs, and functional specs that make delivery predictable — then we build against the same plan.

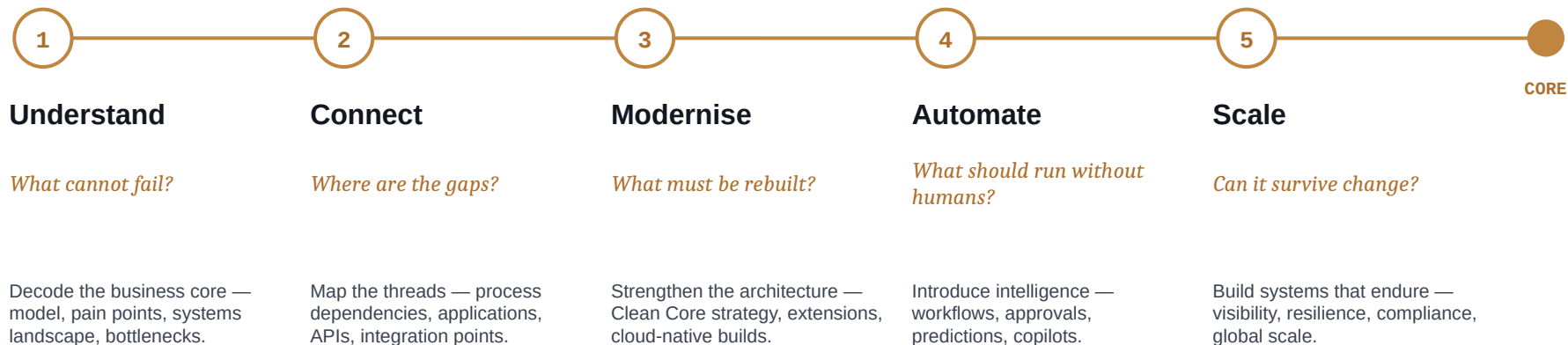
ECOSYSTEM STUDY · DPR · FUNCTIONAL SPECS

*The architect who scopes your project is the one who builds it — senior involvement from first conversation to production.*



# One continuous method, applied at any scale.

Threads are what moves — processes, applications, data, integrations, intelligence, people. The Core is what anchors — value, resilience, governance.





# Fourteen years of shipping enterprise systems.

Delivered through senior architect roles for global enterprises and independent product builds — the experience this practice is built on.

**14+**

years of enterprise delivery

**3**

countries served onsite — India, Germany, Australia

**10+**

enterprise systems architected and shipped

**4 / 7**

regions / languages in one global rollout

**2**

SAP certifications — CAP & Fiori development

**2**

SAP courses taught as instructor — BTP100, CLD200

**INDUSTRIES** Mining & Explosives · Manufacturing · Oil & Gas · Engineering · Healthcare · Supply Chain · HR & Talent · Consumer AI

**STACK** SAP BTP · CAP · Fiori/UI5 · S/4HANA · HANA Cloud · AI Core · Joule · SAP Build · Integration Suite · SAC · React · Node.js · MongoDB · Python · Langchain



# From paper dockets to a connected field operation.

**A global explosives company — Australia · 4 regions · 7 languages · live in production.** Blast schedules lived in spreadsheets. Delivery dockets were paper, signed at remote mine sites. Goods receipt took 16 manual steps. Invoicing waited 3–5 days per docket.



## PLAN

### Planner Web App

A real-time blast scheduling board — trucks, crews, shotfirers, surveyors, multi-day and recurring bookings. An AI-driven 4-week rolling forecast auto-creates Planned Orders in S/4HANA.



## EXECUTE

### Operator Tablet App

Offline-capable digital dockets for remote sites — planned vs actual quantities, on-site additions, customer signature on glass, a legally complete PDF, and an instant invoice trigger.



## BILL & TRACK

### Customer Portal

Mining customers see every delivery live, countersign dockets digitally to release invoicing, and track contract utilisation — the 3–5 day billing delay is gone.

**16 → 1**

GOODS RECEIPT STEPS

**3–5 days → instant    4 · 7**

INVOICE RELEASE

REGIONS · LANGUAGES

**80+**

DEVELOPMENT OBJECTS





# A clean-core, side-by-side extension — end to end.

## PRESENTATION

### FOI Web App

Scheduling · forecast · inbound delivery · stock — React

### Operator Docket App

Offline MDK app — dockets, actuals, signatures

### Customer Portal

Delivery tracking · docket sign-off · invoices — React

OData V4 / REST · MDK offline sync · AI calls via BTP

## SAP BTP RUNTIME & SERVICES

### CAP Backend

CDS entities · OData V4 · Node.js handlers · audit trail · variance detection

### SAP AI Core + Joule Studio

4-week demand forecast model · confidence scoring · natural-language inventory queries

### Mobile Services

MDK runtime · offline store sync · push dispatch notifications

### XSUAA + Destination

OAuth 2.0 · principal propagation — user identity forwarded to S/4HANA

remote OData services · event-driven cascade on docket submit

## DATA & INTEGRATION

### HANA Cloud

Custom entities · calculation views for stock runway & consumption

### S/4HANA — released APIs

PGR posting · Planned Orders · stock overview · material documents · master data sync

### Events & Notifications

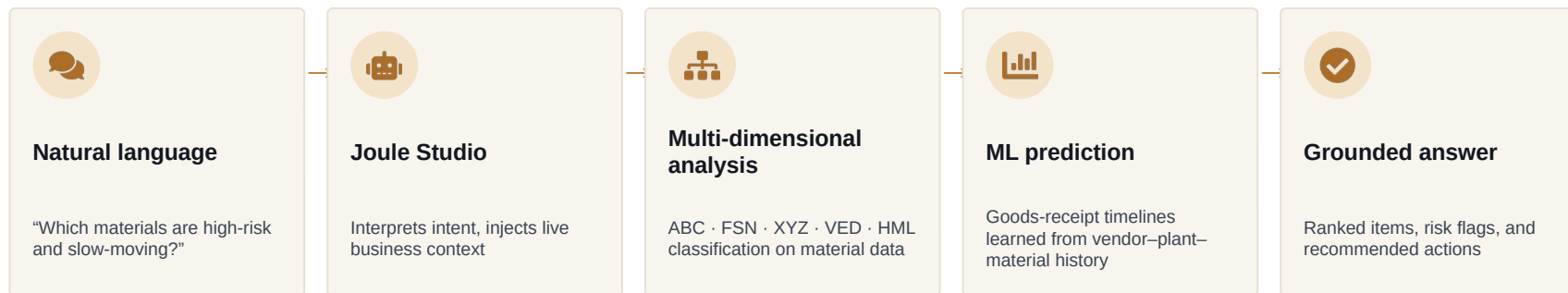
Docket submit → PGR → invoice trigger → customer email + PDF · Twilio SMS dispatch

**Identity-aware by design** — the logged-in user's identity flows from BTP to S/4HANA on every call; the ERP enforces the same authorisations as in the SAP GUI. No shared technical users, no hardcoded credentials.



# Ask your inventory a question. Get a decision.

**AI-Powered Obsolete Inventory Optimizer** — a predictive analytics and GenAI solution on SAP BTP that finds obsolete and high-risk stock, predicts goods-receipt timelines, and answers questions in plain language.

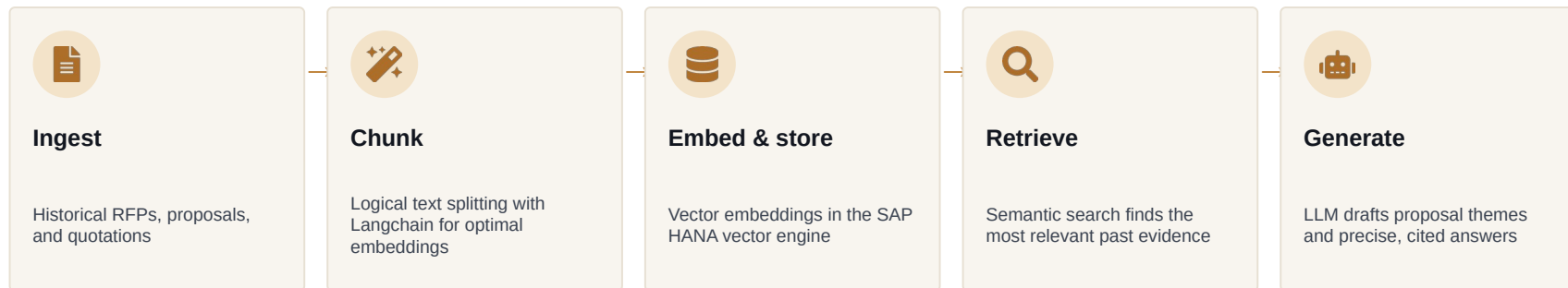


**StockSense AI — supply chain intelligence in the field.** The same Joule + AI Core pattern applied to field execution: mobile stock transfers and approvals, obsolete-material flagging, predictive restock against demand, and conversational queries — "find current stock of 6-inch gate valves at Yard 3" — answered from live data.



# Retrieval-augmented generation, grounded in your data.

**RFP Copilot** — an AI engine that learns from years of past RFPs, quotations, and winning proposals — then drafts context-aware responses and answers bid questions with evidence.

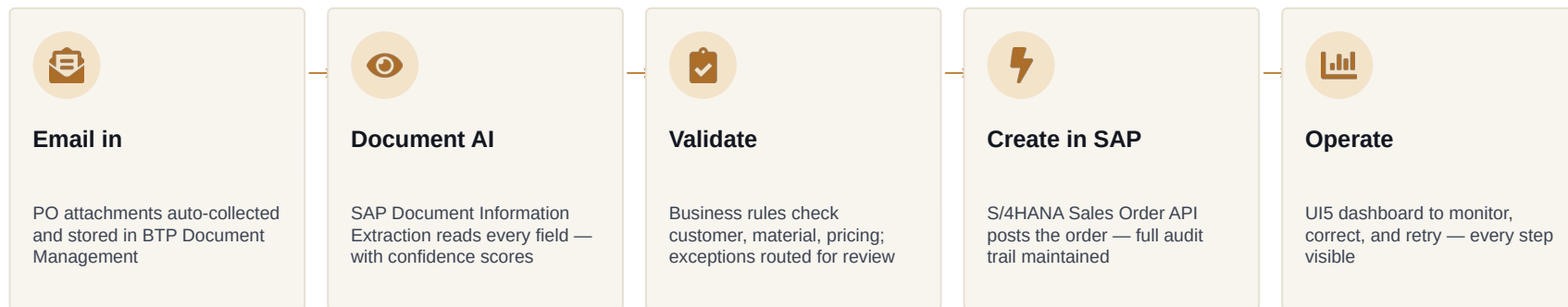


**In use:** “What's the deadline? What should our winning bid value be? What are the key bidder expectations?” — answered in seconds from the document base. **The same RAG pattern powers AyurVaidya's plain-English inventory assistant and CareFlow's clinical summaries.**



## A purchase order arrives by email. A sales order appears in SAP.

**PO-to-SO Automation Pipeline** — an end-to-end document automation flow on SAP BTP that turns inbound customer POs into validated S/4HANA sales orders, with humans only where judgment is needed.



### 90%

LESS MANUAL DATA ENTRY

### hours → minutes

ORDER PROCESSING TIME

### 100%

AUDITABILITY — EVERY CORRECTION LOGGED

SAP BPA · SAP BUILD · DOCUMENT INFORMATION EXTRACTION · BTP DMS · CAP · UI5 · S/4HANA · ODATA



# 45,000 custom objects. One migration answer.

**Clean Core Assessment Dashboard** — global engineering firm, Germany. Analysed 15+ S/4HANA and ECC systems to answer the question every CIO asks before a migration: what do we actually run, and what will it cost us to move?



## Extensibility

Every custom object classified into clean-core tiers 1–3 — cloud-ready, at-risk, or rebuild — to prioritise standardisation.



## Data quality

Completeness, accuracy, and consistency scored per data domain — customer, material, vendor — surfacing weak links before they break a migration.



## Integrations

Standard OData vs legacy RFC / IDoc dependency mapping — the true integration debt made visible.



## Technical debt

ABAP dumps, failed jobs, transport errors, inactive users — a live monitor of system health, not a one-off audit.

## 45,000+

CUSTOM OBJECTS CLASSIFIED

## 15+

SYSTEMS ANALYSED

## 360°

SYSTEM-HEALTH VIEWS IN SAC

HANA CLOUD CALCULATION VIEWS · SAP ANALYTICS CLOUD · CAP · ABAP EXTRACTORS · FIORI ELEMENTS



## We also ship our own AI products.



HEALTHCARE — US CLINICS

### CareFlow

AI clinic operations alongside existing EMRs — live-slot booking, consultation recording to structured SOAP notes in 60 seconds, a follow-up engine that watches for cancellations and offers earlier slots, and controlled medication refills.

FOLLOW-UP COMPLIANCE &lt;40% → 80%+



AYUSH INSTITUTIONS — INDIA

### AyurVaidya

OPEX + CAPEX inventory for Ayurveda hospitals — FEFO batch management, expiry and AMC alerts, QR-coded audit trail, an AI assistant, and anomaly detection that flags unusual consumption spikes and at-risk batches before they become losses. Deployed at the All India Institute of Ayurveda, New Delhi.

125 ITEMS · 28 LIVE ALERTS · ANOMALY FLAGS



CONSUMER AI — FITNESS

### Comeback

AI coaching platform with 11 LLM touchpoints — contextual memory, planned-vs-actual pattern analysis, and vision-based nutrition tracking built for the Indian context.

11 LLM TOUCHPOINTS · PATTERN ANALYSIS

A veterinary platform (PawPilot) ships on the same component architecture — 60% development-effort reuse. Stack: React · Node.js · MongoDB · GenAI · Speech-to-Text · Vision.



## Prashant Agarwal — Founder & Principal Architect

SAP-certified BTP Solution Architect with 14+ years of enterprise delivery — SAP BTP, Clean Core extensions, AI-driven systems, cloud architecture, full-stack engineering, and production-scale applications. He thinks like a business leader and delivers like an engineer.



### CREDENTIALS

SAP Certified Associate — Backend Developer (CAP) · SAP Certified Associate — Fiori Application Developer · SAP Instructor for BTP100 & CLD200 · Corporate SAP BTP training delivered for global enterprises · Industries: mining, manufacturing, oil & gas, engineering, healthcare, HR, analytics



# Start with one problem worth connecting.

A good first engagement is specific enough to deliver value quickly — and important enough to reveal the larger architecture.



2–3 WEEKS

## Architecture Diagnostic

Assessment of systems, process gaps, integration risk, and roadmap options — with a written, actionable output.



4–8 WEEKS

## Focused Pilot

One workflow, integration, dashboard, AI assistant, or ERP extension — shipped to real users.



PROGRAMME

## Modernisation Roadmap

A phased plan for Clean Core, platform adoption, automation, and data — sequenced by business value.



ONGOING

## Build Partnership

Founder-led implementation of enterprise applications, workflows, integrations, and AI-enabled systems.

Engagements run remote or onsite, globally — anchored from Pune, India.





# Tell us how your business runs. We'll tell you what to build.

*We are not taking every engagement. We are taking the right ones — where architecture can create real leverage and the client wants a peer, not a vendor.*

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